

crease in energy capacity is possible.

Cellphones to augment reality

Smartphones will soon add digital features to the world you see around you. Augmenting the reality requires them to draw on nearly every facility they possess—camera, GPS, tilt sensors, digital compass and wireless broadband—to determine exactly what is being looked at, and to find and display relevant extra data. So you will be able to point your phone at an object and see an enhanced version of reality on its screen—a mountain labeled with its height or a person tagged with his name.

The Enkin application developed by two German researchers for phones using Google's Android operating system is a good example. It acts like an enhanced satnav system, helping you to find nearby services or points of interest. Nokia's Mobile Augmented Reality Applications project something similar, drawing its annotations from known points of interest in the satnav software found in Nokia smartphones. Both these systems are under development.

Another, the Wikitude AR Travel Guide, is already available for Android phones. By drawing on Wikipedia it can label sites in the phone's view. Wikitude can also tap into location-based contents of Qype, an online user-based review service, allowing users to stroll down a street and read reviews of restaurants, bars or hotels just by pointing the phone's camera at them.

Disk capable of storing 300 DVDs

A new optical recording method could lead to the development of data disks with 300 times the storage capacity of standard DVDs. This could mean a whopping 1.6 terabytes of information fitted on a DVD-sized disk.

The technique, named 'five-dimensional' optical recording, employs nanometre-scale particles of gold as a recording medium. Researchers at Swinburne University of Technology in Australia exploited the particular

properties of these gold 'nano-rods' by manipulating the light pointed at them. They were able to add three 'dimensions' to the two spatial dimensions that DVDs and CDs already have—a spectral (colour) dimension and a polarisation dimension, as well as recording information in ten layers of the nano-rod films, adding a third spatial dimension.

The scientists used the nanoparticles to record information in a range of different colour wavelengths on the same physical disc location. This is a major improvement over traditional DVDs, which are recorded in a single colour wavelength with a laser.

Since the amount of incoming laser light absorbed by the nanoparticles depends on its polarisation, the researchers were able to record different layers of information at different angles.

Previously demonstrated recording techniques were based on colour or polarisation, but this is the first work that integrates both. As a result, the scientists have achieved unprecedented data density.

Computer that can recognise languages

Scientists in England have developed lip-reading computers that can distinguish between different languages. The discovery could be useful for deaf people, for law enforcement agencies and in noisy environments. The research was led by Stephen Cox and Jake Newman of University of East Anglia (UEA).

In order to develop the technology, the lip motions made by a group of 23 bilingual and trilingual speakers were statistically modeled. The system was able to accurately identify which language was spoken by an individual speaker. These languages included English, French, German, Arabic, Mandarin, Cantonese, Italian, Polish and Russian.

The research is part of a wider UEA project on automatic lip-reading. The next step will be to make the system more robust to an individual's physiology and his way of speaking.

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